Surface Mount Schottky Power Rectifier

POWERMITE®
Power Surface Mount Package

MBRM140T1G, NRVBM140T1G, MBRM140T3G, NRVBM140T3G

The Schottky POWERMITE® employs the Schottky Barrier principle with a barrier metal and epitaxial construction that produces optimal forward voltage drop-reverse current tradeoff. The advanced packaging techniques provide for a highly efficient micro miniature, space saving surface mount Rectifier. With its unique heatsink design, the POWERMITE® has the same thermal performance as the SMA while being 50% smaller in footprint area, and delivering one of the lowest height profiles, < 1.1 mm in the industry. Because of its small size, it is ideal for use in portable and battery powered products such as cellular and cordless phones, chargers, notebook computers, printers, PDAs and PCMCIA cards. Typical applications are AC-DC and DC-DC converters, reverse battery protection, and "ORing" of multiple supply voltages and any other application where performance and size are critical.

Features

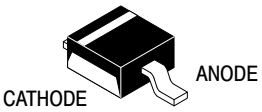
- Low Profile-Maximum Height of 1.1 mm
- Small Footprint-Footprint Area of 8.45 mm^2
- Low V_F Provides Higher Efficiency and Extends Battery Life
- Supplied in 12 mm Tape and Reel
- Low Thermal Resistance with Direct Thermal Path of Die on Exposed Cathode Heat Sink
- ESD Ratings:
  - ♦ Human Body Model = 3B (> 16000 V)
  - ♦ Machine Model = C (> 400 V)
- AEC–Q101 Qualified and PPAP Capable
- NRVB Prefix for Automotive and Other Applications Requiring Unique Site and Control Change Requirements
- All Packages are Pb-Free*

Mechanical Characteristics:

- POWERMITE® is JEDEC Registered as D0-216AA
- Case: Molded Epoxy
- Epoxy Meets UL 94 V-0 @ 0.125 in
- Weight: 16.3 mg (Approximately)
- Lead and Mounting Surface Temperature for Soldering Purposes: 260 °C Maximum for 10 Seconds

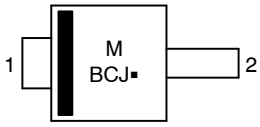
*For additional information on our Pb-Free strategy and soldering details, please download the **onsemi** Soldering and Mounting Techniques Reference Manual, SOLDERRM/D

SCHOTTKY BARRIER RECTIFIER 1.0 AMPERES, 40 VOLTS



POWERMITE
CASE 457
PLASTIC

MARKING DIAGRAM



M = Date Code
BCJ = Device Code
▪ = Pb-Free Package

ORDERING INFORMATION

Device	Package	Shipping†
MBRM140T1G	POWERMITE (Pb-Free)	3,000 / Tape & Reel
NRVBM140T1G	POWERMITE (Pb-Free)	3,000 / Tape & Reel
MBRM140T3G	POWERMITE (Pb-Free)	12,000 / Tape & Reel

DISCONTINUED (Note 1)

NRVBM140T3G	POWERMITE (Pb-Free)	12,000 / Tape & Reel

† For information on tape and reel specifications, including part orientation and tape sizes, please refer to our Tape and Reel Packaging Specifications Brochure, BRD8011/D.

1. **DISCONTINUED:** This device is not available. Please contact your **onsemi** representative for information. The most current information on this device may be available on www.onsemi.com.

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MAXIMUM RATINGS

Rating	Symbol	Value	Unit
Peak Repetitive Reverse Voltage Working Peak Reverse Voltage DC Blocking Voltage	V_{RRM} V_{RWM} V_R	40	V
Average Rectified Forward Current (At Rated V_R, T_C = 110 °C)	I_O	1.0	A
Peak Repetitive Forward Current (At Rated V_R, Square Wave, 100 kHz, T_C = 110 °C)	I_{FRM}	2.0	A
Non-Repetitive Peak Surge Current (Non-Repetitive peak surge current, halfwave, single phase, 60 Hz)	I_{FSM}	50	A
Storage Temperature	T_{stg}	−55 to 150	°C
Operating Junction Temperature	T_J	−55 to 125	°C
Voltage Rate of Change (Rated V_R, T_J = 25 °C)	dv/dt	10,000	V/μs

Stresses exceeding Maximum Ratings may damage the device. Maximum Ratings are stress ratings only. Functional operation above the Recommended Operating Conditions is not implied. Extended exposure to stresses above the Recommended Operating Conditions may affect device reliability.

THERMAL CHARACTERISTICS

Characteristic	Symbol	Value	Unit
Thermal Resistance, Junction-to-Lead (Anode) (Note 1) Thermal Resistance, Junction-to-Tab (Cathode) (Note 1) Thermal Resistance, Junction-to-Ambient (Note 1)	R_{tjl} R_{tjtab} R_{tja}	35 23 277	°C/W

1. Mounted with minimum recommended pad size, PC Board FR4, See Figures 9 & 10

ELECTRICAL CHARACTERISTICS

Characteristic	Symbol	Value		Unit
Maximum Instantaneous Forward Voltage (Note 2), See Figure 2	V_F	**T_J = 25°C**	**T_J = 85°C**	V
(I_F = 0.1 A) (I_F = 1.0 A) (I_F = 3.0 A)		0.36 0.55 0.85	0.30 0.515 0.88	
Maximum Instantaneous Reverse Current (Note 2), See Figure 4	I_R	**T_J = 25°C**	**T_J = 85°C**	mA
(V_R = 40 V) (V_R = 20 V)		0.5 0.15	25 18	

2. Pulse Test: Pulse Width ≤ 250 μs, Duty Cycle ≤ 2%

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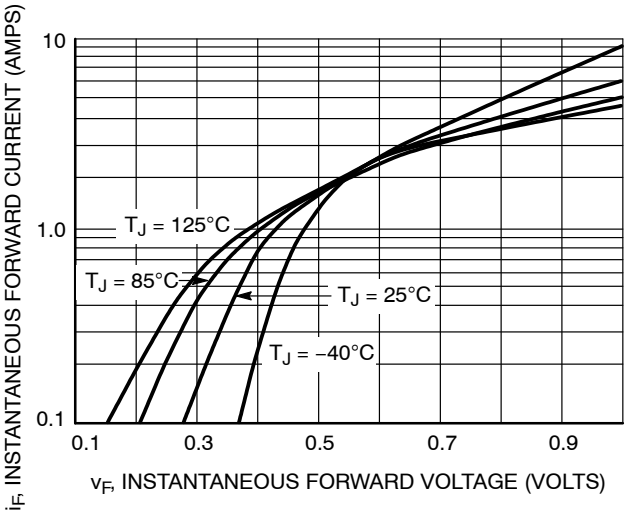


Figure 1. Typical Forward Voltage

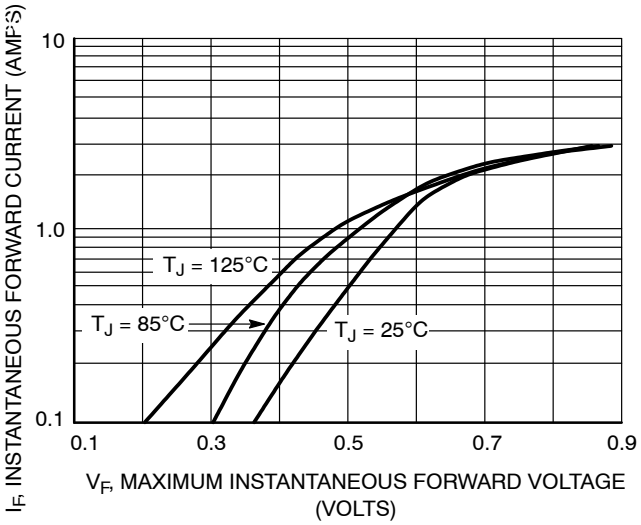


Figure 2. Maximum Forward Voltage

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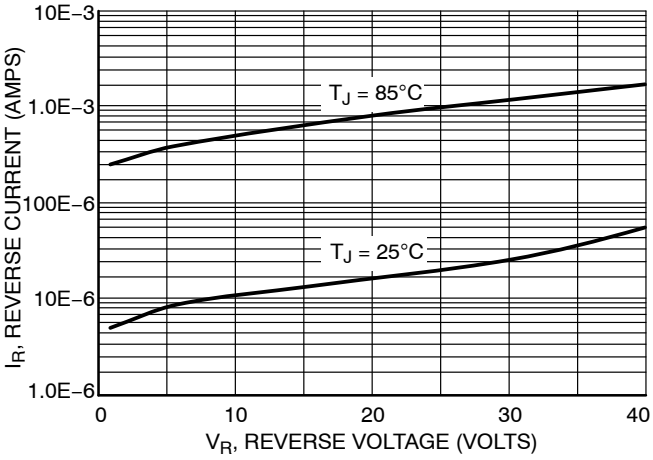


Figure 3. Typical Reverse Current

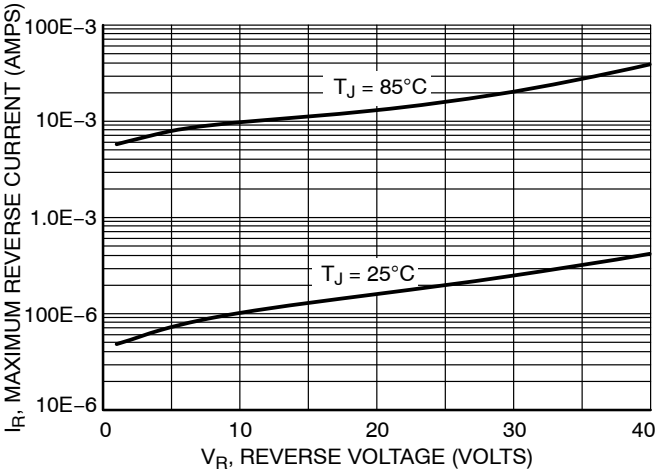


Figure 4. Maximum Reverse Current

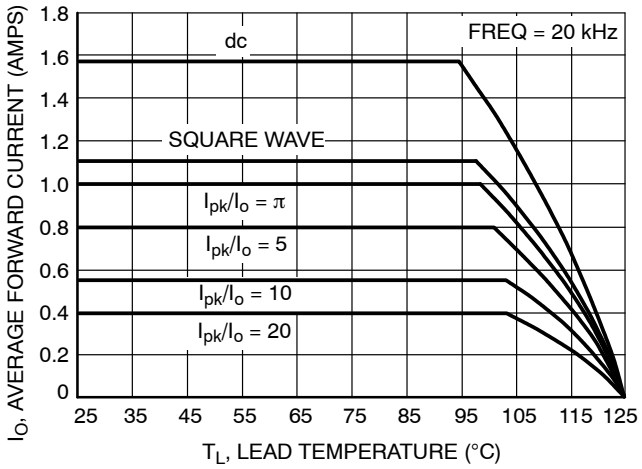


Figure 5. Current Derating

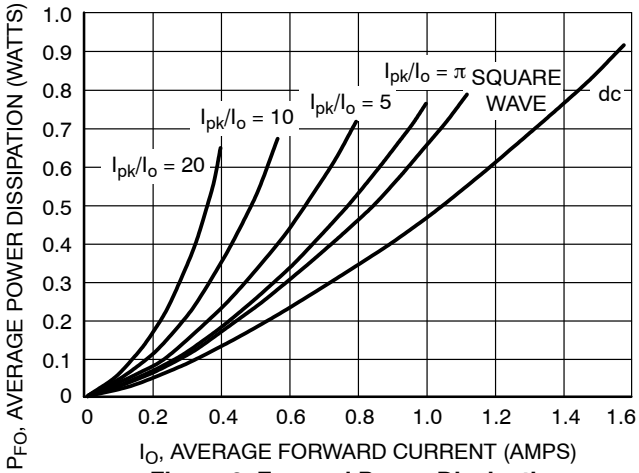


Figure 6. Forward Power Dissipation

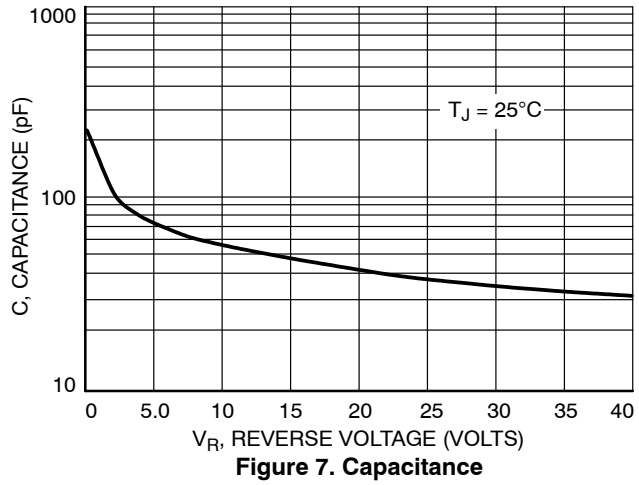


Figure 7. Capacitance

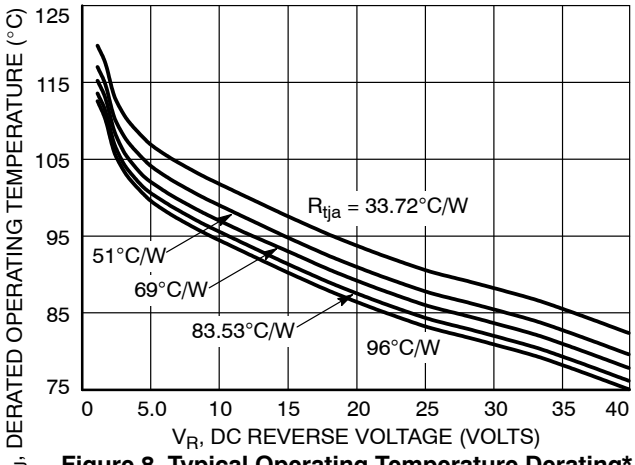


Figure 8. Typical Operating Temperature Derating*

* Reverse power dissipation and the possibility of thermal runaway must be considered when operating this device under any reverse voltage conditions. Calculations of T_J therefore must include forward and reverse power effects. The allowable operating T_J may be calculated from the equation: $T_J = T_{Jmax} - r(t)(Pf + Pr)$ where

$r(t)$ = thermal impedance under given conditions,
Pf = forward power dissipation, and
Pr = reverse power dissipation

This graph displays the derated allowable T_J due to reverse bias under DC conditions only and is calculated as $T_J = T_{Jmax} - r(t)Pr$, where r(t) = Rthja. For other power applications further calculations must be performed.

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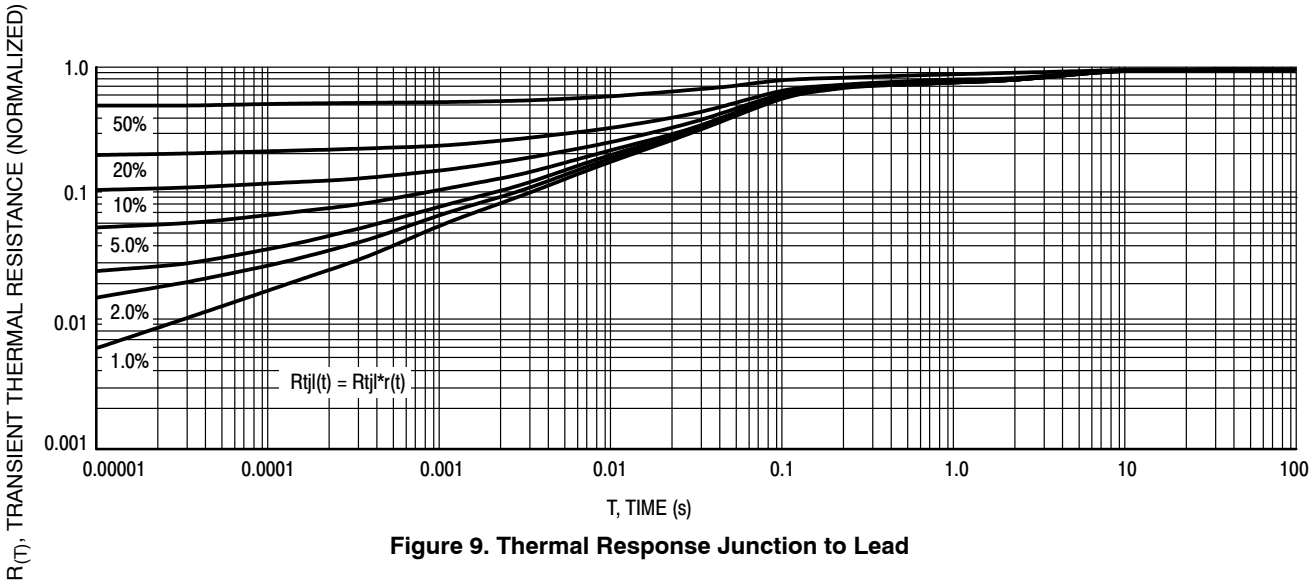


Figure 9. Thermal Response Junction to Lead

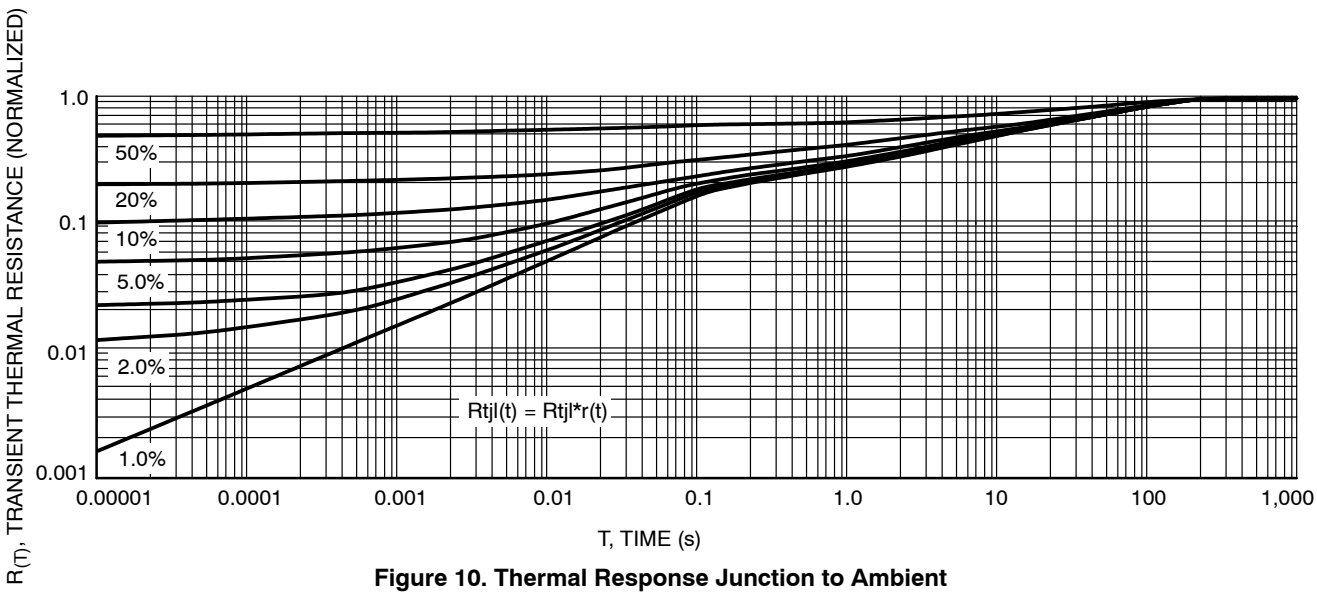


Figure 10. Thermal Response Junction to Ambient

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REVISION HISTORY

Revision	Description of Changes	Date
5	NRVBM140T3G OPN Marked as Discontinued + Rebranded the Data Sheet to **onsemi** format	7/4/2025

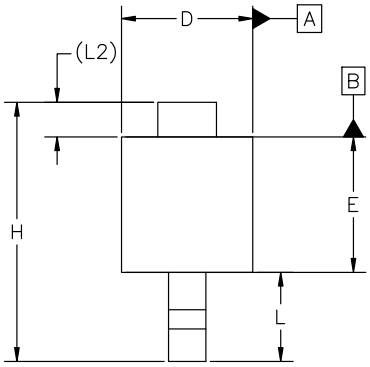
This document has undergone updates prior to the inclusion of this revision history table. The changes tracked here only reflect updates made on the noted approval dates.

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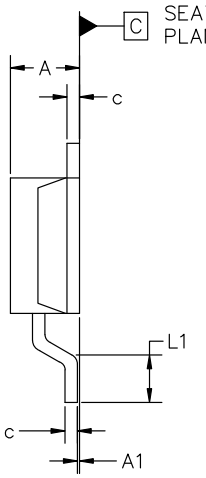


POWERMITE 1.90x1.96x1.00
CASE 457
ISSUE H

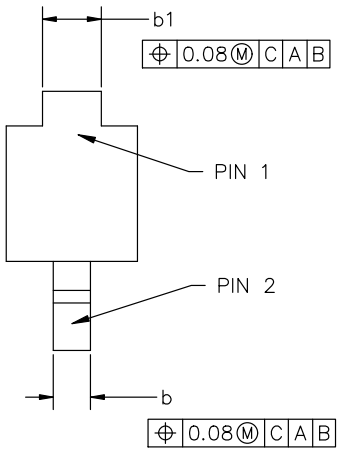
DATE 16 MAY 2025



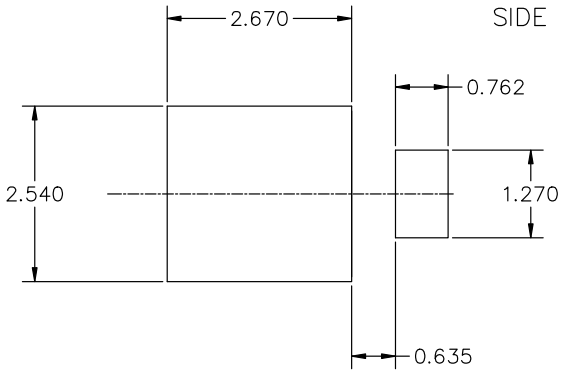
TOP VIEW



SIDE VIEW



BOTTOM VIEW



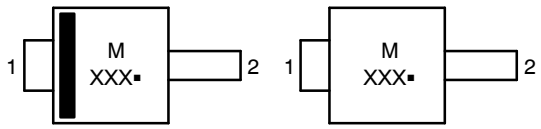
RECOMMENDED MOUNTING FOOTPRINT

*For additional information on our Pb−Free Strategy and Soldering and Mounting Techniques Reference Manual, SOLDERRM/D.

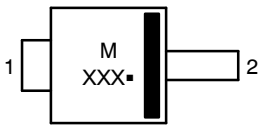
MILLIMETERS			
DIM	MIN	NOM	MAX
A	0.85	1.00	1.15
A1	0.00	0.05	0.10
b	0.40	0.55	0.69
b1	0.70	0.85	1.00
c	0.10	0.18	0.25
D	1.75	1.90	2.05
E	1.75	1.96	2.18
H	3.60	3.75	3.90
L	1.20	1.35	1.50
L1	0.50	0.65	0.80
L2	0.50 REF		

NOTES:
1. DIMENSIONING AND TOLERANCING AS PER ASME Y14.5M, 2018.
2. CONTROLLING DIMENSION: MILLIMETERS.
3. DIMENSION D & E DO NOT INCLUDE MOLD FLASH, PROTRUSIONS OR GATE BURRS.

GENERIC MARKING DIAGRAMS*



STYLE 1 **STYLE 2**



STYLE 3

XXX = Specific Device Code
M = Date Code
▪ = Pb−Free Package

STYLE 1:
PIN 1. CATHODE
2. ANODE

STYLE 2:
PIN 1. ANODE OR CATHODE
2. CATHODE OR ANODE
(BI−DIRECTIONAL)

STYLE 3:
PIN 1. ANODE
2. CATHODE

*This information is generic. Please refer to device data sheet for actual part marking. Pb−Free indicator, “G” or microdot “▪”, may or may not be present. Some products may not follow the Generic Marking.

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DESCRIPTION:	**POWERMITE 1.90x1.96x1.00**		**PAGE 1 OF 1**

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